| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 12(a) | $\begin{pmatrix} 10 \\ -15 \end{pmatrix}$ | **1** | |
| 12(b) | 10 | **4** | **B3** for answer ±10<br>OR<br>**M3** for $(-4-4)^2 + (0-6)^2$ oe or better<br>OR<br>**M2** for $\begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 6 \end{pmatrix}$ oe<br>or **B1** for [$J$ =] (4, 6) soi or $\begin{pmatrix} -1 \\ -8 \end{pmatrix}$<br>**M1** for $(\textit{their} - 8)^2 + (\textit{their} - 6)^2$ oe or better |
| 13(a) | 120<br>and<br>opposite angles of a cyclic quadrilateral sum to 180° | **2** | **B1** for 120 or a fully correct reason |
| 13(b) | 25 | **2** | **M1** for (180 – 120) – 35<br>or **B1** for $ABC = 60$ or $ABD = 35$ |
| 14 | $90x^4y^3$ final answer | **2** | **B1** for two correct parts from:<br>90, $x^4$ and $y^3$<br>or for $3\times5\times x\times y\times y\times y$ and<br>$2\times3\times3\times x\times x\times x\times x\times y$<br>or for correct answer seen then spoilt |
| 15(a) | $5\sqrt{3}$ | **2** | **B1** for $3\sqrt{3}$ or $2\sqrt{3}$ |
| 15(b) | 16 | **2** | **M1** for $40\times2\sqrt{2}$ , $8\times2\sqrt{2}$ , $\frac{8\sqrt{4}\sqrt{2}}{\sqrt{2}}$ or $\frac{40\sqrt{8}\sqrt{2}}{5\sqrt{2}\sqrt{2}}$ or better<br>or **B1** for $8\sqrt{4}$ or $\sqrt{256}$ |
| 15(c) | $\frac{3+\sqrt{5}}{4}$ | **2** | **M1** for $\frac{1}{3-\sqrt{5}}\times\frac{3+\sqrt{5}}{3+\sqrt{5}}$ oe |